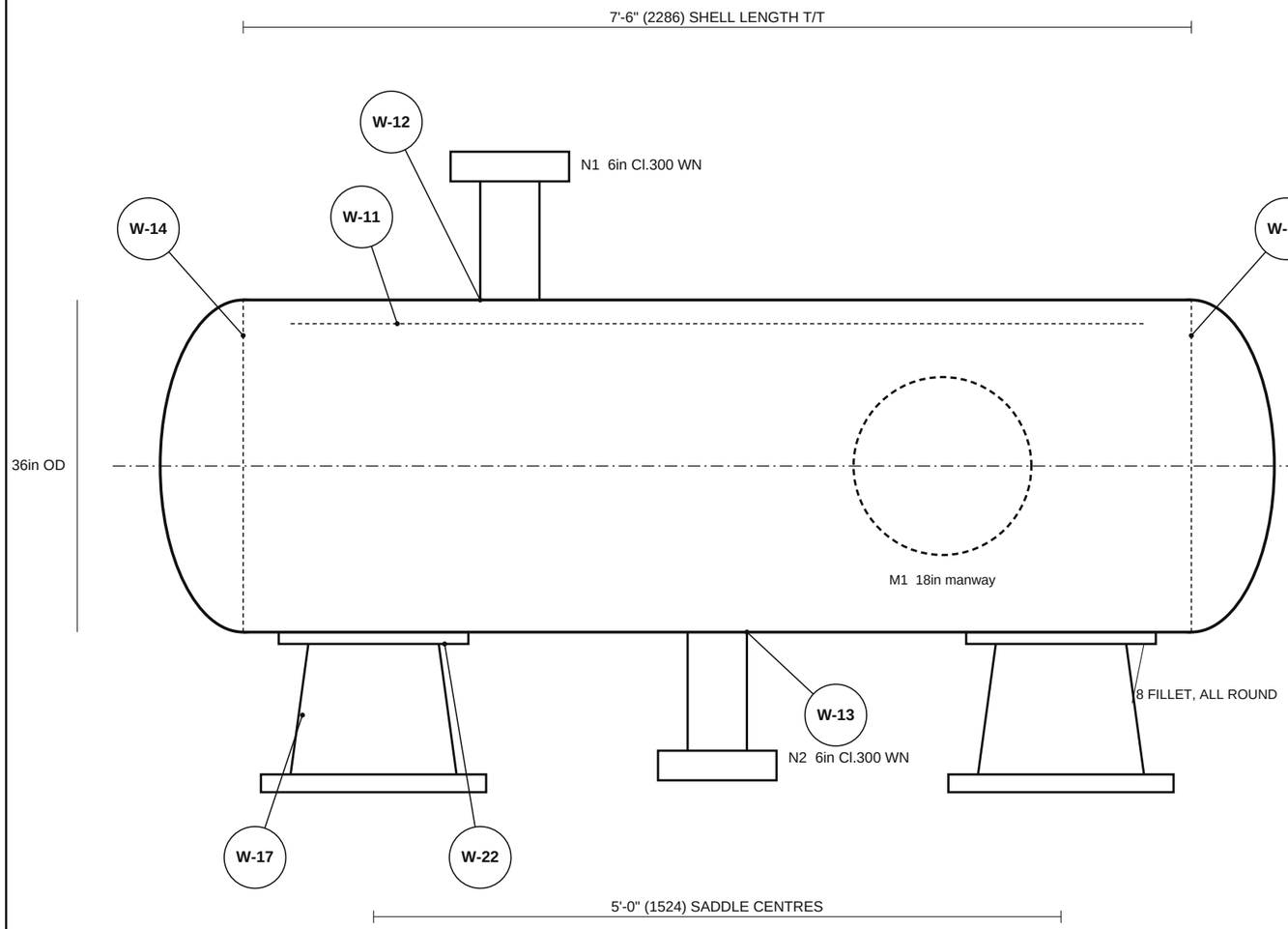
SYNTHETIC DOCUMENT - generated for software testing - not a real drawing

ELEVATION · NOT TO SCALE



WELD MAP / WELD TABLE

Joint	Description	Cat.	Type	t (mm)	Pos.	WPS	Welder	NDE	PWHT
W-11	Shell long seam	A	1	40	3G	WPS-07	118	RT 100%	Yes
W-12	Nozzle N1 to shell	D	7	40	6G	WPS-07	118	UT 100% + RT 10%	Yes
W-13	Nozzle N2 to shell	D	7	40	6G	WPS-07	118	UT 100%	Yes
W-14	Shell to head, left (circ)	B	1	40	2G	WPS-07	118	RT 100%	Yes
W-15	Shell to head, right (circ)	B	1	40	2G	WPS-07	118	RT 100%	Yes
W-17	Saddle to wear plate, fillet	-	fillet	12	2F	WPS-09	204	VT 100%	No
W-22	Wear plate to shell, fillet	-	fillet	12	4F	WPS-09	204	VT 100% + MT 10%	No

MATERIAL LIST

Item	Part	Material	t (mm)	Heat no.
1	Shell, 1 course	SA-516 Gr.70 N	40	4A7721
2	Heads, 2:1 SE (2)	SA-516 Gr.70 N	40	4A7721
3	Nozzle neck N1, N2 6in	SA-106 Gr.B	SCH 80	5B2210
4	Flanges N1, N2 6in Cl.300 WN	SA-105	-	5C1180
5	Saddles and wear plates	S355J2+N	12	3C0954

DESIGN DATA

Code	ASME BPVC VIII-1, 2023 edition
Design pressure	1.6 MPa (16 bar)
Design temperature	210 °C
MDMT	-20 °C at 1.6 MPa
Corrosion allowance	3 mm
Joint efficiency	E = 1.0 (full RT, Cat. A and B)
Hydrotest	2.08 MPa, 30 min (UG-99)
PWHT	Required, UCS-56: 620 ± 15 °C, 1 h per 25 mm

NOTES

1. Construction per ASME BPVC VIII-1, 2023 edition. Welding per ASME IX; NDE per ASME V.
2. All Category A and B butt welds: full radiography per UW-11(a).
3. PWHT after all pressure welding is complete; no welding after PWHT.
4. Weld IDs in circles refer to the weld table. Production positions as listed.
5. Saddle and wear-plate welds are non-pressure attachments.

PRESSURE VESSEL V-110 · SHOP DRAWING / WELD MAP

Drawing no. **HBK-PV-110**	Revision **B**
Client **Northfield Refining (fictional)**	Scale **NTS**
Drawn **M. Tanaka (fictional) 2026-09-18**	Checked **R. Lindqvist (fictional)**